Optimization 1
ISE 5406

Lecture 2: Linear Algebra, Convex Sets & Convex Functions

Dr. Robert Hildebrand

Grado Department of Industrial & Systems Engineering
Virginia Tech

Where We Are in the Course

Last Time (Lecture 1): Introduction to Nonlinear Programming

- Course overview and applications of nonlinear programming
- Mathematical modeling and optimization formulations
- Why NLP is harder than LP: local vs. global optima
- Fundamentals of unconstrained optimization

Today (Lecture 2): Introduction to Convexity

- Linear algebra review: combinations, independence, span
- Convex sets: definitions, examples, operations preserving convexity
- Convex and concave functions
- Convex hulls and Carathéodory's theorem
- Introduction to convex analysis: extreme points, recession directions

Looking Ahead

Next Time (Lecture 3): Real Analysis and Separation

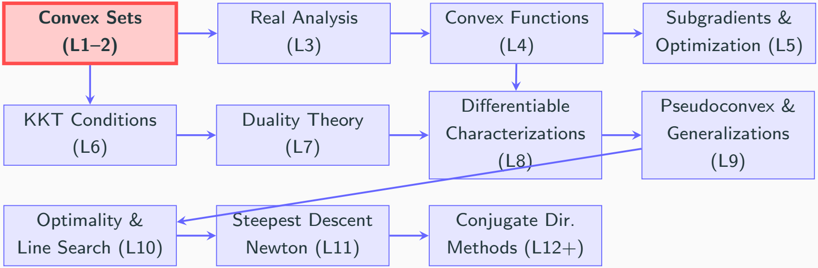
- Closed, open, and compact sets
- Existence of minimum/maximum: Weierstrass' theorem
- Separating hyperplane theorem
- Supporting hyperplane theorem

Coming Soon (Lecture 4): More on Convex Functions

- Strictly convex and strictly concave functions
- Operations preserving convexity of functions
- Continuity of convex functions
- Epigraphs and hypographs

Big Picture: Lectures 2–4 build the convex analysis foundation needed for optimality conditions and algorithms later in the course.

Big Picture: Course Roadmap



Why Convexity Matters

- Convex problems: Local optimum = Global optimum (tractable!)
- Non-convex problems: Convex analysis still provides key tools
- Foundation for: KKT conditions, duality, convergence analysis

Reading for this lecture

- Chapter 2 of Bazaraa, Sherali, and Shetty (2006)
- Parts of Chapters 2 and 3 of Boyd and Vandenberghe

Outline

Linear Algebra Review

Convex Sets

- Examples of Convex Sets
- Operations Preserving Convexity of Sets

Convex and Concave Functions

- Examples of Convex and Concave Functions

Convex Hulls

- Carathéodory's Theorem

Introduction to Convex Analysis

Roadmap

Linear Algebra Review

Convex Sets

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- Examples of Convex and Concave Functions

Convex Hulls

- Carathéodory's Theorem

Introduction to Convex Analysis

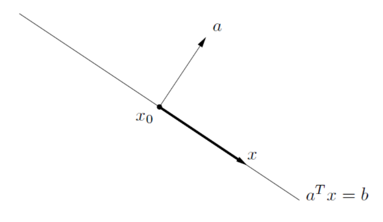
Definitions: Linear Algebra

- Euclidean Space ($\mathbb{R}^n$): The collection of all vectors of dimension n.
- Hyperplane: A set of the form

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T\mathbf{x} = b\},$$

  where $\mathbf{a} \in \mathbb{R}^n \backslash \{\mathbf{0}\}$ and $b \in \mathbb{R}$.

  - A line when $n = 2$ and a plane when $n = 3$.



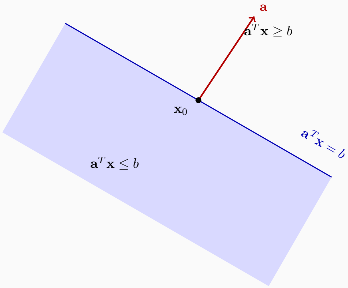
  - Set of points $\mathbf{x}$ with constant inner product (b) to given vector $\mathbf{a}$.
  - A hyperplane divides $\mathbb{R}^n$ into two **halfspaces**.

Linear Algebra: Definitions

- Halfspaces:

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T\mathbf{x} \leq b\} \text{ and } \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T\mathbf{x} \geq b\},$$

where $\mathbf{a} \in \mathbb{R}^n \backslash \{\mathbf{0}\}$ and $b \in \mathbb{R}$.



Linear Algebra: Definitions

Consider vectors $\mathbf{a}_1$, $\mathbf{a}_2, \ldots, \mathbf{a}_k$ in $\mathbb{R}^n$.

- Linear Combination: A vector $\mathbf{b} \in \mathbb{R}^n$ is said to be a *linear* combination of $\mathbf{a}_1$, $\mathbf{a}_2, \ldots, \mathbf{a}_k$, if

$$\mathbf{b} = \sum_{j=1}^{k} \lambda_j \mathbf{a}_j$$

  where $\lambda_j \in \mathbb{R}$ for $j = 1, \ldots, k$.

- Conic Combination: A vector $\mathbf{b} \in \mathbb{R}^n$ is said to be a *conic* combination of $\mathbf{a}_1$, $\mathbf{a}_2, \ldots, \mathbf{a}_k$, if

$$\mathbf{b} = \sum_{j=1}^{k} \lambda_j \mathbf{a}_j$$

  where $\lambda_j \geq 0$ for $j = 1, \ldots, k$.

Linear Algebra: Definitions

Consider vectors $\mathbf{a}_1$, $\mathbf{a}_2, \ldots, \mathbf{a}_k$ in $\mathbb{R}^n$.

- Affine Combination: A vector $\mathbf{b} \in \mathbb{R}^n$ is said to be an *affine* combination of $\mathbf{a}_1$, $\mathbf{a}_2, \ldots, \mathbf{a}_k$, if

$$\mathbf{b} = \sum_{j=1}^{k} \lambda_j \mathbf{a}_j$$

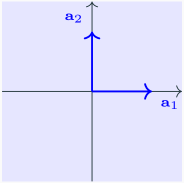
  where $\lambda_j \in \mathbb{R}$ for $j = 1, \ldots, k$, and $\sum_{j=1}^{k} \lambda_j = 1$.

- Convex Combination: A vector $\mathbf{b} \in \mathbb{R}^n$ is said to be a *convex* combination of $\mathbf{a}_1$, $\mathbf{a}_2, \ldots, \mathbf{a}_k$, if

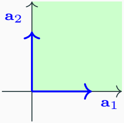
$$\mathbf{b} = \sum_{j=1}^{k} \lambda_j \mathbf{a}_j$$

  where $\lambda_j \in [0, 1]$ for $j = 1, \ldots, k$, and $\sum_{j=1}^{k} \lambda_j = 1$.

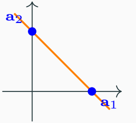
Types of Combinations: Visual Summary



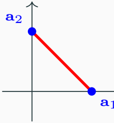
Linear: all $\mathbb{R}^2$



Conic: $\lambda_j \geq 0$



Affine: $\sum \lambda_j = 1$



Convex: $\lambda_j \in [0$

Type	Coefficients	Sum Constraint
Linear	$\lambda_j \in \mathbb{R}$	None
Conic	$\lambda_j \geq 0$	None
Affine	$\lambda_j \in \mathbb{R}$	$\sum \lambda_j = 1$
Convex	$\lambda_j \geq 0$	$\sum \lambda_j = 1$

Linear Algebra: Definitions

- Linear/Affine Subspace ($S_L \subseteq \mathbb{R}^n$): If $\mathbf{a}_1$ and $\mathbf{a}_2 \in S_L$, then every linear/affine combination of $\mathbf{a}_1$ and $\mathbf{a}_2 \in S_L$.
- **Linear Independence:** Vectors $\mathbf{a}_1$, $\mathbf{a}_2, \ldots, \mathbf{a}_k$ in $\mathbb{R}^n$ are said to be linearly independent if

$$\sum_{j=1}^{k} \lambda_j \mathbf{a}_j = \mathbf{0} \implies \lambda_j = 0, \text{ for all } j = 1, \ldots, k.$$

- **Affine Independence:** Vectors $\mathbf{a}_1$, $\mathbf{a}_2, \ldots, \mathbf{a}_k$ in $\mathbb{R}^n$ are said to be affinely independent if $\mathbf{a}_2 - \mathbf{a}_1$, $\mathbf{a}_3 - \mathbf{a}_1, \ldots, \mathbf{a}_k - \mathbf{a}_1$ are linearly independent.

$$\sum_{j=1}^{k} \lambda_j = 0, \; \sum_{j=1}^{k} \lambda_j \mathbf{a}_j = \mathbf{0} \implies \lambda_j = 0, \text{ for all } j = 1, \ldots, k.$$

- **Span:** Vectors $\mathbf{a}_1$, $\mathbf{a}_2, \ldots, \mathbf{a}_k$ in $\mathbb{R}^n$ are said to span $\mathbb{R}^n$ if *any* vector in $\mathbb{R}^n$ can be represented as their linear combination.

Roadmap

Linear Algebra Review

Convex Sets

- Examples of Convex Sets
- Operations Preserving Convexity of Sets

Convex and Concave Functions

- Examples of Convex and Concave Functions

Convex Hulls

- Carathéodory's Theorem

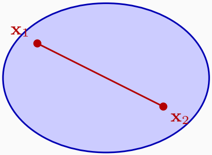
Introduction to Convex Analysis

Convex Sets

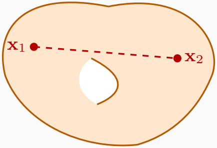
- **Convex Set:** A set $X \subseteq \mathbb{R}^n$ is convex if for *any* two points $\mathbf{x}_1$ and $\mathbf{x}_2$ in X and *any* $\lambda \in [0, 1]$,

$$\lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 \in X.$$

- Geometric Interpretation: For *each* pair of points $\mathbf{x}_1$ and $\mathbf{x}_2$ in X, the line segment joining them must be *entirely* contained in X.



Convex set



Nonconvex set

Convex Sets: Examples

- Which of the following sets are convex?

  1. The Euclidean space $\mathbb{R}^n$ and the empty set $\emptyset$
  2. $\{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$ and $\{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1\}$
  3. $\{\mathbf{x} : A\mathbf{x} = \mathbf{b}\}$, where A is an $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$
  4. $\{\mathbf{x} : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0\}$, where A is an $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$
  5. Hyperplanes and halfspaces
  6. $\{\mathbf{x} \in \mathbb{R}_+^2 : x_1 x_2 = 0\}$
  7. $\{\mathbf{x} \in \mathbb{R}^2 : x_1^2 - x_1 x_2^2 + x_2^4 + 1 \geq 0\}$

Exercise: Convex Combinations

Quick Check

Given vectors $\mathbf{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{a}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ in $\mathbb{R}^2$:

1. Is $\mathbf{b} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$ a convex combination of $\mathbf{a}_1$ and $\mathbf{a}_2$?

2. Is $\mathbf{b} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ a convex combination of $\mathbf{a}_1$ and $\mathbf{a}_2$?

3. What *type* of combination is $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$? (linear, conic, affine, or convex?)

Solution: Convex Combinations

Answers

1. Yes. We need $\lambda_1, \lambda_2 \geq 0$ with $\lambda_1 + \lambda_2 = 1$ such that:

$$\lambda_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$$

   Solution: $\lambda_1 = 0.5$, $\lambda_2 = 0.5$. Both ≥ 0 and sum to 1. $\checkmark$

2. No. We would need $\lambda_1 = 2$ and $\lambda_2 = -1$, but $\lambda_2 < 0$ violates the non-negativity requirement for convex combinations.

3. Affine combination. We have $\lambda_1 + \lambda_2 = 2 + (-1) = 1$, so the sum constraint is satisfied, but $\lambda_2 = -1 < 0$, so it's not convex or conic. It is a linear combination (any λ's work) and an affine combination (sum = 1).

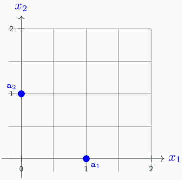
Exercise: Geometric Interpretation of Combinations

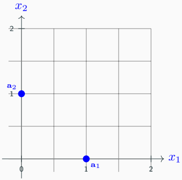
Think-Pair-Share

Consider vectors $\mathbf{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{a}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ in $\mathbb{R}^2$.

Sketch the set of *all* possible:

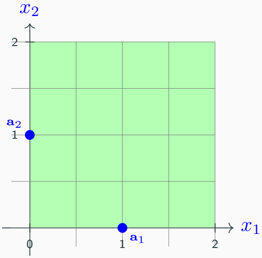
1. **Conic combinations** of $\mathbf{a}_1$ and $\mathbf{a}_2$
2. **Convex combinations** of $\mathbf{a}_1$ and $\mathbf{a}_2$



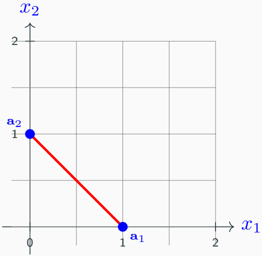


Solution: Geometric Interpretation of Combinations

Answers



Conic: First quadrant $\mathbb{R}^2_+$



Convex: Line segment

- Conic: All points $\lambda_1\mathbf{a}_1 + \lambda_2\mathbf{a}_2$ with $\lambda_1, \lambda_2 \geq 0$ = entire first quadrant
- Convex: Same but with $\lambda_1 + \lambda_2 = 1$ = line segment from $\mathbf{a}_1$ to $\mathbf{a}_2$

Examples of Convex Sets: Polyhedra and Polyhedral Cones

Polyhedral Set or Polyhedron: Intersection of a finite number of halfspaces; $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \leq \mathbf{b}\}$, where $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$.

- Polytope: A bounded polyhedron.
- Polyhedral sets are convex.
- (Geometrically) redundant constraint: Polyhedral set is not affected by removing this constraint (no new points are added).

Polyhedral Cone: Intersection of a finite number of halfspaces, whose hyperplanes pass through the origin, i.e., $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \leq \mathbf{0}\}$.

- Special case of polyhedral sets.

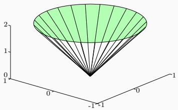
Convex Cone: A convex set X with the additional property

$$\mathbf{x} \in X \text{ and } \lambda \geq 0 \implies \lambda\mathbf{x} \in X.$$

Polyhedral cones are a special case of convex cones.

Examples of Convex Sets: Ellipsoids and Convex Cones

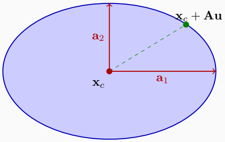
Examples of convex cones:



- Lorentz or Second-Order Cone: $\mathcal{L}^{n+1} := \{(\mathbf{x}, t) \in \mathbb{R}^n \times \mathbb{R} : \|\mathbf{x}\|_2 \leq t\}$.
- Positive Semidefinite Cone: $\mathcal{S}_+^n := \{X \in \mathcal{S}^n : X \succeq 0\}$, where $\mathcal{S}^n$ is the set of symmetric $n \times n$ matrices.

Other examples of convex sets:

- **Euclidean Ball:** $\mathcal{B}(\mathbf{x}_c, r) := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_c\|_2 \leq r\}$.
- **Ellipsoid:** $\mathcal{E} := \{\mathbf{x}_c + A\mathbf{u} : \|\mathbf{u}\|_2 \leq 1\} \subseteq \mathbb{R}^n$, where $A \in \mathcal{S}_+^n$.



$\mathcal{E} = \{\mathbf{x}_c + \mathbf{Au} : \|\mathbf{u}\|_2 \leq 1\}$

Operations Preserving Convexity of Sets (Not Exhaustive)

Lemma
Let S, S_1, S_2 be convex sets in $\mathbb{R}^n$. Then, the following sets are convex:

1. Intersection: $S_1 \cap S_2$.
2. Minkowski Addition: $S_1 \oplus S_2 := \{\mathbf{x}_1 + \mathbf{x}_2 : \mathbf{x}_1 \in S_1, \mathbf{x}_2 \in S_2\}$.
3. Minkowski Difference: $S_1 \ominus S_2 := \{\mathbf{x}_1 - \mathbf{x}_2 : \mathbf{x}_1 \in S_1, \mathbf{x}_2 \in S_2\}$.
4. Translation: $S + a := \{\mathbf{x} + a : \mathbf{x} \in S\}$ for *any* $a \in \mathbb{R}^n$.
5. Scaling: $tS := \{t\mathbf{x} : \mathbf{x} \in S\}$ for *any* $t \in \mathbb{R}$.
6. Cartesian Product: $S_1 \times S_2 := \{(\mathbf{x}_1, \mathbf{x}_2) : \mathbf{x}_1 \in S_1, \mathbf{x}_2 \in S_2\}$.
7. Coordinate Projection: $\text{proj}_{\mathbf{x}}(S) := \{\mathbf{x} : (\mathbf{x}, \mathbf{y}) \in S \text{ for } \textit{some } \mathbf{y}\}$.
8. Affine Image: $\{A\mathbf{x} + \mathbf{b} : \mathbf{x} \in S\}$ for *any* $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$.
9. Affine Pre-image: $\{\mathbf{x} : A\mathbf{x} + \mathbf{b} \in S\}$ for *some* $A \in \mathbb{R}^{n \times m}$, $\mathbf{b} \in \mathbb{R}^n$.

Exercise: Proving Minkowski Sum Preserves Convexity

Group Exercise

Let S_1 and S_2 be convex sets in $\mathbb{R}^n$. Prove that the Minkowski sum

$$S_1 \oplus S_2 := \{\mathbf{x}_1 + \mathbf{x}_2 : \mathbf{x}_1 \in S_1, \mathbf{x}_2 \in S_2\}$$

is convex.

Proof:

Hint: Take two points $\mathbf{z}_1 = \mathbf{x}_1 + \mathbf{y}_1$ and $\mathbf{z}_2 = \mathbf{x}_2 + \mathbf{y}_2$ in $S_1 \oplus S_2$, and show their convex combination is also in $S_1 \oplus S_2$.

Solution: Proving Minkowski Sum Preserves Convexity

Proof

Let $\mathbf{z}_1, \mathbf{z}_2 \in S_1 \oplus S_2$. Then:

- $\mathbf{z}_1 = \mathbf{x}_1 + \mathbf{y}_1$ for some $\mathbf{x}_1 \in S_1$, $\mathbf{y}_1 \in S_2$
- $\mathbf{z}_2 = \mathbf{x}_2 + \mathbf{y}_2$ for some $\mathbf{x}_2 \in S_1$, $\mathbf{y}_2 \in S_2$

For any $\lambda \in [0, 1]$:

$$\begin{aligned}
\lambda \mathbf{z}_1 + (1-\lambda)\mathbf{z}_2 &= \lambda(\mathbf{x}_1 + \mathbf{y}_1) + (1-\lambda)(\mathbf{x}_2 + \mathbf{y}_2) \\
&= \underbrace{[\lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2]}_{\in S_1 \text{ (convexity)}} + \underbrace{[\lambda \mathbf{y}_1 + (1-\lambda)\mathbf{y}_2]}_{\in S_2 \text{ (convexity)}}
\end{aligned}$$

Since S_1 is convex: $\lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2 \in S_1$

Since S_2 is convex: $\lambda \mathbf{y}_1 + (1-\lambda)\mathbf{y}_2 \in S_2$

Therefore, $\lambda \mathbf{z}_1 + (1-\lambda)\mathbf{z}_2 \in S_1 \oplus S_2$. $\square$

Roadmap

Linear Algebra Review

Convex Sets

- Examples of Convex Sets
- Operations Preserving Convexity of Sets

Convex and Concave Functions

- Examples of Convex and Concave Functions

Convex Hulls

- Carathéodory's Theorem

Introduction to Convex Analysis

Convex and Concave Functions

Let $X \subseteq \mathbb{R}^n$ be a convex set.

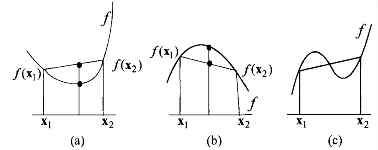
Convex Function: A function $f : X \to \mathbb{R}$ is said to be convex if for *any* two points $\mathbf{x}_1$ and $\mathbf{x}_2$ in X and *any* $\lambda \in [0,1]$,

$$f(\lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1-\lambda) f(\mathbf{x}_2).$$

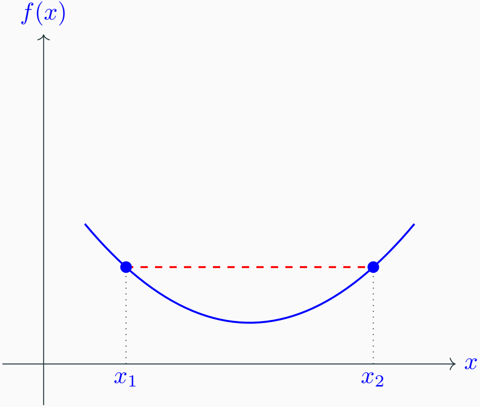
Concave Function: A function $f : X \to \mathbb{R}$ is said to be concave if $-f$ is convex, i.e., for *any* two points $\mathbf{x}_1$ and $\mathbf{x}_2$ in X and *any* $\lambda \in [0,1]$,

$$f(\lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2) \geq \lambda f(\mathbf{x}_1) + (1-\lambda) f(\mathbf{x}_2).$$

Only affine functions ($\mathbf{a}^T\mathbf{x} + b$) are *both* convex and concave.



Convex Function: Geometric Interpretation



Key insight: For a convex function, the graph lies *below* any chord connecting two points.

Exercise: Classifying Functions

Classification Challenge (2 minutes)

For each function below, determine if it is **convex**, **concave**, or **neither**:

Function	Answer
$f(x) = \|x\|$ on $\mathbb{R}$	
$f(x) = \sin(x)$ on $[0, \pi]$	
$f(\mathbf{x}) = \max(x_1, x_2)$ on $\mathbb{R}^2$	
$f(\mathbf{x}) = x_1 x_2$ on $\mathbb{R}^2$	
$f(\mathbf{x}) = x_1 x_2$ on $\mathbb{R}^2_{++}$	

Hint: Think about the second derivative test for univariate functions, or try specific examples.

Solution: Classifying Functions

Answers

Function	Answer
$f(x) = \|x\|$ on $\mathbb{R}$	Convex (V-shape, epigraph is convex)
$f(x) = \sin(x)$ on $[0, \pi]$	Concave ($f'' = -\sin(x) \leq 0$)
$f(\mathbf{x}) = \max(x_1, x_2)$ on $\mathbb{R}^2$	Convex (pointwise max of affine)
$f(\mathbf{x}) = x_1 x_2$ on $\mathbb{R}^2$	Neither (Hessian indefinite)
$f(\mathbf{x}) = x_1 x_2$ on $\mathbb{R}^2_{++}$	Neither (still indefinite)

Note on $f(\mathbf{x}) = x_1 x_2$: The Hessian is $H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ with eigenvalues ± 1, so it's indefinite (neither PSD nor NSD).

Exercise: Proving Convexity from Definition

Group Exercise

Prove that $f(x) = x^2$ is convex on $\mathbb{R}$ using the **definition** of convexity.

Goal: Show that for any $x_1, x_2 \in \mathbb{R}$ and $\lambda \in [0, 1]$:

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

Proof:

Hint: Expand both sides and show LHS − RHS ≤ 0.

Solution: Proving Convexity from Definition

Proof that $f(x) = x^2$ is convex

LHS: $f(\lambda x_1 + (1-\lambda)x_2) = (\lambda x_1 + (1-\lambda)x_2)^2$

$$= \lambda^2 x_1^2 + 2\lambda(1-\lambda)x_1x_2 + (1-\lambda)^2 x_2^2$$

RHS: $\lambda f(x_1) + (1-\lambda)f(x_2) = \lambda x_1^2 + (1-\lambda)x_2^2$

Difference (RHS − LHS):

$$\begin{aligned}
&= \lambda x_1^2 + (1-\lambda)x_2^2 - \lambda^2 x_1^2 - 2\lambda(1-\lambda)x_1x_2 - (1-\lambda)^2 x_2^2 \\
&= \lambda(1-\lambda)x_1^2 - 2\lambda(1-\lambda)x_1x_2 + \lambda(1-\lambda)x_2^2 \\
&= \lambda(1-\lambda)(x_1^2 - 2x_1x_2 + x_2^2) \\
&= \lambda(1-\lambda)(x_1 - x_2)^2 \geq 0
\end{aligned}$$

Since RHS − LHS ≥ 0, we have LHS $\leq$ RHS. $\square$

Examples of Convex and Concave Functions

Examples of convex functions:

- Exponential function: $\exp(x)$ on $X = \mathbb{R}$.
- Power functions: x^p on $X = (0, +\infty)$ for $p \geq 1$ or $p \leq 0$.
- Negative entropy: $x \ln(x)$ on $X = (0, +\infty)$.
- Quadratic functions: $\mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ on $X = \mathbb{R}^n$ for any $A \in \mathcal{S}_+^n$, $\mathbf{b} \in \mathbb{R}^n$, $c \in \mathbb{R}$.

Examples of concave functions:

- Power functions: x^p on $X = (0, +\infty)$ for $p \in [0, 1]$.
- Geometric mean: $\left(\prod_{i=1}^n x_i\right)^{1/n}$ on $X = \mathbb{R}_{++}^n$.
- Logarithm: $\ln(x)$ on $X = (0, +\infty)$.
- Quadratic functions: $\mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ on $X = \mathbb{R}^n$ for $-A \in \mathcal{S}_+^n$, $\mathbf{b} \in \mathbb{R}^n$, $c \in \mathbb{R}$.

Level Sets of Convex Functions are Convex Sets

(Lower) Level Set: Let S be a convex set and $f : S \to \mathbb{R}$ be a convex function. Then, for any $\alpha \in \mathbb{R}$, the α-level set of f is defined as

$$S_\alpha = \{\mathbf{x} \in S : f(\mathbf{x}) \leq \alpha\}.$$

Lemma
The set S_α is a convex set.

Proof: Let $\mathbf{x}_1, \mathbf{x}_2 \in S_\alpha$, so $f(\mathbf{x}_1) \leq \alpha$ and $f(\mathbf{x}_2) \leq \alpha$.

For $\lambda \in [0,1]$: $f(\lambda\mathbf{x}_1 + (1-\lambda)\mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1-\lambda) f(\mathbf{x}_2) \leq \alpha$.

- **Upper** level sets of concave functions are convex.
- Suppose $g_i : \mathbb{R}^n \to \mathbb{R}$, $i = 1, \ldots, m_I$, are convex and $h_i : \mathbb{R}^n \to \mathbb{R}$, $i = 1, \ldots, m_E$, are affine. Then,

  $$\{\mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) \leq 0,\ i = 1, \ldots, m_I,\ \ h_j(\mathbf{x}) = 0,\ j = 1, \ldots, m_E\}$$

  is a convex set (why?)

Roadmap

Linear Algebra Review

Convex Sets

- Examples of Convex Sets
- Operations Preserving Convexity of Sets

Convex and Concave Functions

- Examples of Convex and Concave Functions

Convex Hulls

- Carathéodory's Theorem

Introduction to Convex Analysis

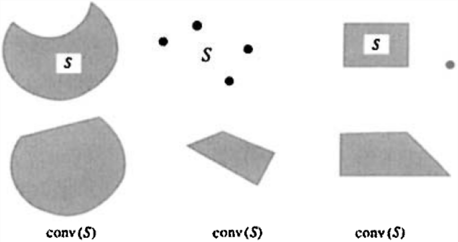
Convex Hulls

Definition

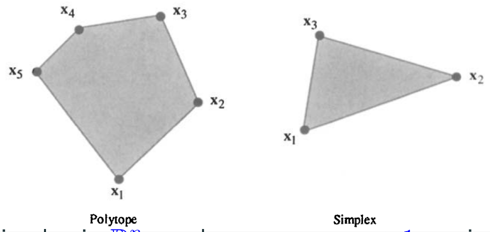
- Let S be an **arbitrary** set in $\mathbb{R}^n$. The convex hull of S, denoted conv(S), is the set of all convex combinations of S.
- In other words, $\mathbf{x} \in$ conv(S) if and only if it can be represented as

$$\mathbf{x} = \sum_{j=1}^{k} \lambda_j \mathbf{x}_j, \quad \sum_{j=1}^{k} \lambda_j = 1, \quad \lambda_j \geq 0, \; j = 1, \ldots k,$$

for some positive integer k and points $\mathbf{x}_1, \ldots, \mathbf{x}_k \in S$.



- Let $S := \{\mathbf{x}_1, \ldots, \mathbf{x}_{k+1}\}$ be a finite set of points in $\mathbb{R}^n$.
- **Polytope:** Convex hull of S.
- **Simplex:** If $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{k+1}$ are affinely independent, then conv(S) is called a simplex having vertices $\mathbf{x}_1, \ldots, \mathbf{x}_{k+1}$.



- Note: Any simplex in $\mathbb{R}^n$ can have at most $n+1$ vertices (why?)

Lemma

Let S be an arbitrary set in $\mathbb{R}^n$. Then, conv(S) is the smallest convex set containing S. It is also the intersection of all convex sets containing S.

Carathéodory's Theorem (Convex Hull)

Carathéodory's Theorem
Let S be an arbitrary set in $\mathbb{R}^n$. If $\mathbf{x} \in \mathrm{conv}(S)$, then

- there exist $\mathbf{x}_1, \ldots, \mathbf{x}_{n+1} \in S$ such that $\mathbf{x} \in \mathrm{conv}(\mathbf{x}_1, \ldots, \mathbf{x}_{n+1})$.
- In other words, there exist $\mathbf{x}_1, \ldots, \mathbf{x}_{n+1} \in S$ such that

$$\mathbf{x} = \sum_{j=1}^{n+1} \lambda_j \mathbf{x}_j, \quad \sum_{j=1}^{n+1} \lambda_j = 1, \quad \lambda_j \geq 0, \; j = 1, \ldots n+1,$$

Proof: reading assignment (see Theorem 2.1.6 of Bazaraa et al.)

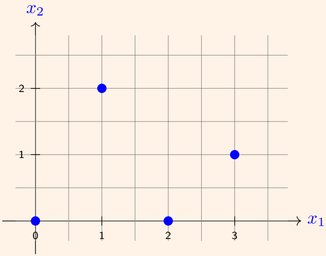
- Trivially holds for $\mathbf{x} \in S$.
- The choice of $\mathbf{x}_1, \ldots, \mathbf{x}_{n+1}$ can depend on $\mathbf{x}$.

Exercise: Convex Hulls

Geometric Construction

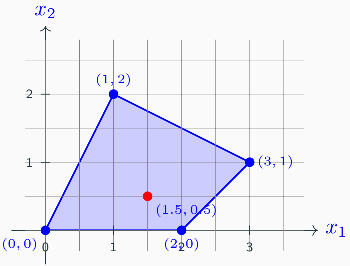
Consider the set $S = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \end{pmatrix} \right\}$ in $\mathbb{R}^2$.

1. Sketch $\text{conv}(S)$ on the grid below.
2. Which points in S are **extreme points** of $\text{conv}(S)$?
3. Is $\begin{pmatrix} 1.5 \\ 0.5 \end{pmatrix}$ in $\text{conv}(S)$? If so, write it as a convex combination.



Solution: Convex Hulls

Answers



1. The convex hull is the quadrilateral shown (all 4 points are vertices).
2. All four points are extreme points (corners of the polygon).
3. Yes. $(1.5, 0.5) = \frac{1}{2}(0, 0) + \frac{1}{2}(3, 1)$ since $\frac{1}{2}(0) + \frac{1}{2}(3) = 1.5$ and $\frac{1}{2}(0) + \frac{1}{2}(1) = 0.5$. ✓

Roadmap

Linear Algebra Review

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Convex and Concave Functions

- Examples of Convex and Concave Functions

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Introduction to Convex Analysis

Convex Analysis: Definitions

Extreme Point: Let X be a convex set. A point $\mathbf{x} \in X$ is called an extreme point of X if it *cannot* be represented as a *strict convex combination* of two distinct points in X.

- Mathematically, if $\mathbf{x} = \lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2$ for some $\lambda \in (0,1)$ and $\mathbf{x}_1, \mathbf{x}_2 \in X$, then $\mathbf{x} = \mathbf{x}_1 = \mathbf{x}_2$.

Ray: A collection of points of the form $\{\mathbf{x}_0 + \lambda \mathbf{d} : \lambda \geq 0\}$ where $\mathbf{d} \neq \mathbf{0}$.

- Vertex of the ray: $\mathbf{x}_0$
- Direction of the ray: $\mathbf{d}$
- Is a ray a convex set? Yes!

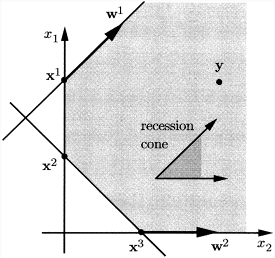
Convex Analysis: Definitions

Recession Directions of a Convex Set: Let X be a convex set. If for *each* $\mathbf{x}_0 \in X$, the ray

$$\{\mathbf{x}_0 + \lambda \mathbf{d} : \lambda \geq 0\} \subseteq X$$

then $\mathbf{d} \neq \mathbf{0}$ is called a recession direction of X.

Example: $X := \{\mathbf{x} \in \mathbb{R}^2_+ : x_1 - x_2 \geq -2, x_1 + x_2 \geq 1\}$



Exercise: Extreme Points

Conceptual Question

Consider the set $X = \{\mathbf{x} \in \mathbb{R}^2 : x_1 \geq 0,\ x_2 \geq 0,\ x_1 + x_2 \leq 1\}$ (a triangle).

1. Is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ an extreme point of X? **Prove your answer.**
2. List *all* extreme points of X.
3. Can $(0.5, 0.25)^T$ be written as a strict convex combination of two distinct points in X?

Solution: Extreme Points

Answers

1. Yes, $(0, 0)$ is an extreme point.

   Proof: Suppose $(0, 0) = \lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2$ for $\lambda \in (0, 1)$ and $\mathbf{x}_1, \mathbf{x}_2 \in X$.

   Since $x_1 \geq 0$ and $x_2 \geq 0$ for all points in X, and the combination equals $(0, 0)$, we need $\lambda x_1^{(1)} + (1 - \lambda)x_1^{(2)} = 0$ with all terms ≥ 0.

   This implies $x_1^{(1)} = x_1^{(2)} = 0$. Similarly $x_2^{(1)} = x_2^{(2)} = 0$.

   Therefore $\mathbf{x}_1 = \mathbf{x}_2 = (0, 0)$. ✓

2. Three extreme points: $(0, 0)$, $(1, 0)$, $(0, 1)$ — the vertices of the triangle.

3. Yes. For example: $(0.5, 0.25) = 0.5 \cdot (0, 0.5) + 0.5 \cdot (1, 0)$.

   Since $(0.5, 0.25)$ is not a vertex, it can be written as a strict convex combination.

Exercise: Recession Directions

Recession Direction Exercise

For the set $X = \{\mathbf{x} \in \mathbb{R}^2_+ : x_1 - x_2 \geq -2\}$:

1. Identify *all* recession directions of X.

2. Describe the recession cone (the set of all recession directions plus $\mathbf{0}$).

3. Is $\mathbf{d} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ a recession direction? Why or why not?

Solution: Recession Directions

Answers

1. Recession directions must satisfy: starting from any $\mathbf{x} \in X$, moving in direction $\mathbf{d}$ keeps us in X.

   For $X = \{\mathbf{x} \in \mathbb{R}^2_+ : x_1 - x_2 \geq -2\}$, we need:

   - $d_1 \geq 0$ (to stay in $x_1 \geq 0$)
   - $d_2 \geq 0$ (to stay in $x_2 \geq 0$)
   - $d_1 - d_2 \geq 0$ (to preserve $x_1 - x_2 \geq -2$)

2. Recession cone: $\{\mathbf{d} \in \mathbb{R}^2 : d_1 \geq 0, d_2 \geq 0, d_1 \geq d_2\}$

   This is the region between the positive x_1-axis and the line $d_1 = d_2$.

3. No. $\mathbf{d} = (1, 2)^T$ has $d_1 - d_2 = 1 - 2 = -1 < 0$.

   This violates the constraint $d_1 \geq d_2$, so it's not a recession direction.

End-of-Lecture Assessment

Quick Assessment (5 questions)

Answer True or False, or provide a short answer:

1. **True/False:** Every affine subspace is convex.
2. **True/False:** The intersection of infinitely many convex sets is always convex.
3. Find all extreme points of the set $\{\mathbf{x} \in \mathbb{R}^2 : |x_1| + |x_2| \leq 1\}$.
4. Is $f(x, y) = xy$ convex on $\mathbb{R}^2$?
5. **True/False:** The projection of a convex set onto a subspace is convex.

End-of-Lecture Assessment: Solutions

Answers

1. True. An affine subspace is a translated linear subspace, and linear subspaces are convex. Translation preserves convexity.
2. True. The intersection of any collection (finite or infinite) of convex sets is convex.
3. Four extreme points: $(1, 0)$, $(-1, 0)$, $(0, 1)$, $(0, -1)$.

   This is the diamond (rotated square) with vertices on the axes.
4. No. The Hessian $H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has eigenvalues ± 1, so it's indefinite. The function is neither convex nor concave.
5. True. Projection is a linear transformation, and affine images of convex sets are convex.

Summary and Next Steps

Key Takeaways

- Convex sets: line segment between any two points stays in the set
- Convex functions: function lies below the chord
- Operations preserving convexity: intersection, affine images, etc.
- Convex hulls and Carathéodory's theorem
- Extreme points, rays, and recession directions

Next Time:

- More on convex analysis: supporting hyperplanes, separation theorems
- Characterizations of convex functions

Questions?